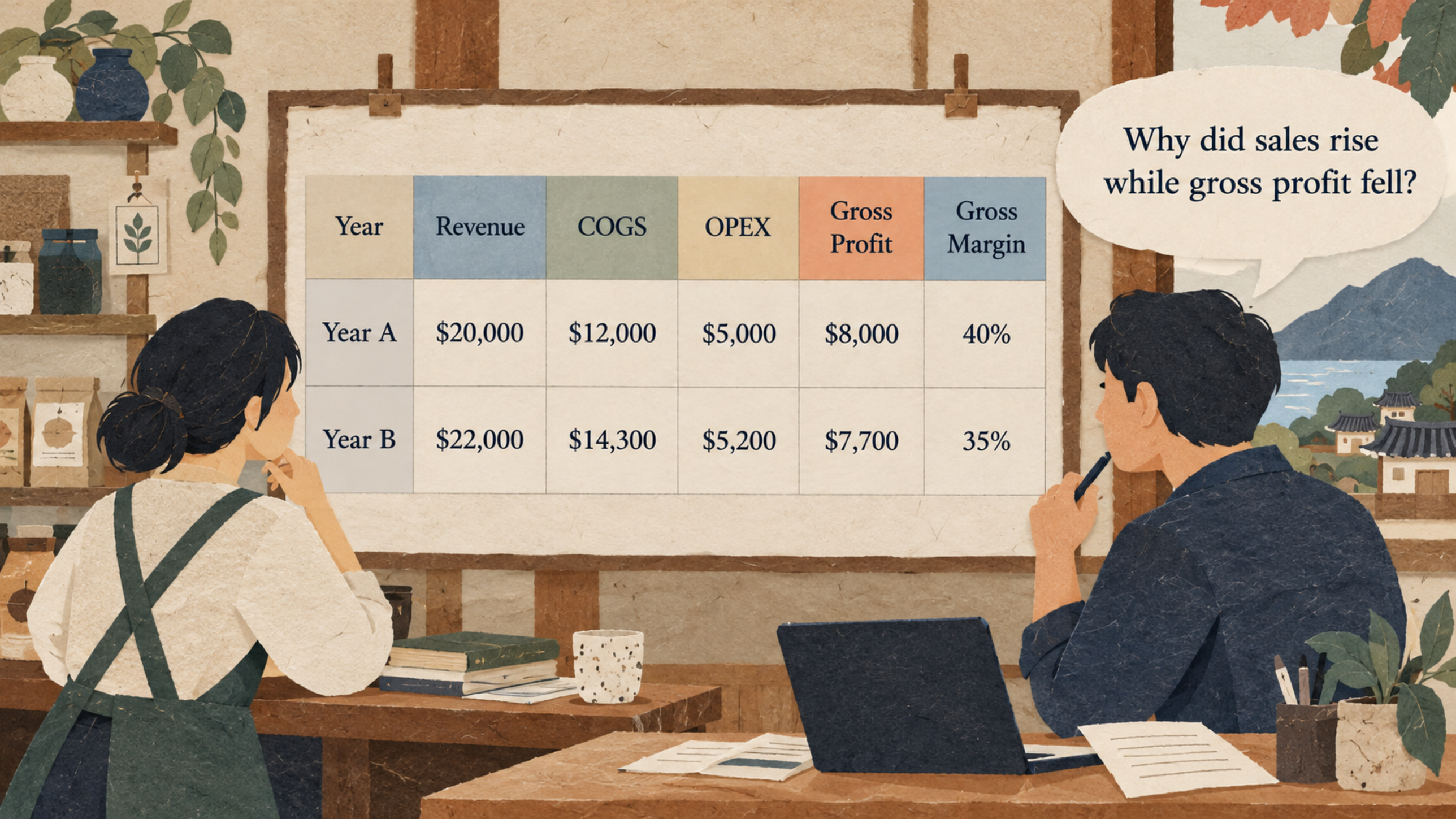


Year	Revenue	COGS	OPEX	Gross Profit	Gross Margin
Year A	$20,000	$12,000	$5,000	$8,000	40%
Year B	$22,000	$14,300	$5,200	$7,700	35%

What problem are we solving?

3 hours: 50-minute lesson, 10-minute break, 50-minute guided practice, 10-minute break, 50-minute applied practice, and 10-minute wrap-up.

SITUATION

The problem

Sales can rise while the money left after direct costs falls. A manager needs to explain the change without confusing sales, direct costs, and overhead.

TARGET

Learning goal

Calculate gross profit and gross margin, check the arithmetic, and write a short evidence-based briefing.

Practice material: Use the practice material supplied in the session Markdown.

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Practice data (fictional shop)

- Year A: revenue $20,000; cost of goods sold (COGS) $12,000; operating expenses (OPEX) $5,000. Year B: revenue $22,000; COGS $14,300; OPEX $5,200. COGS is the direct cost of goods sold; OPEX is the cost of running the business outside those direct costs.

BOUNDARY

Use the supplied facts

Sales can rise while the money left after direct costs falls. A manager needs to explain the change without confusing sales, direct costs, and overhead.

Understand the idea and apply it

Work through the explanation before comparing prompts or starting the exercise.

LEARN

Keep revenue, gross profit, and margin distinct

- Revenue is sales before costs. Gross profit = revenue - cost of goods sold (COGS), the direct cost of the items sold. Gross margin = gross profit / revenue x 100. Margin describes the share of revenue left after direct product costs; it is not net profit because operating expenses are still outside this calculation.

LEARN

Work through the supplied two-year figures

- Year A gross profit is $20,000 - $12,000 = $8,000 and margin is $8,000 / $20,000 = 40%. Year B gross profit is $22,000 - $14,300 = $7,700 and margin is $7,700 / $22,000 = 35%. Revenue rose by $2,000, while gross profit fell by $300 and gross margin fell by 5 percentage points. Verify each formula in a spreadsheet or calculator before explaining the result.

Understand the idea and apply it

Work through the explanation before comparing prompts or starting the exercise.

LEARN

Explain the change without inventing a cause

- The figures establish what changed, but not why. Higher direct costs may be worth investigating, but these four figures do not identify which cost, product, or decision caused the change. Label an explanation as a hypothesis and ask for the next useful evidence, such as item-level costs or sales mix. A short manager brief should separate checked facts from the follow-up question.

APPLY

Three-step method

- Calculate each year's gross profit and margin; compare the figures and label facts versus possible explanations; write a brief that names one question to investigate next.

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

```
Is this shop doing well? Tell me what to do.
```

AVOID

Weak example 2

```
Prove that higher sales mean the business is more profitable.
```

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

```
Using only the four figures for Year A and Year B
below, calculate gross profit
and gross margin. Show the formula and
arithmetic. Do not call gross profit net profit.
```

USE

Improved example 2

```
Use these checked results: Year A gross profit
$8,000 and margin 40%;
Year B gross profit $7,700 and margin 35%. Write
a 100-word manager brief:
state two facts, one possible explanation labeled
as unverified, and one follow-up question.
Do not invent a cause.
```

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Calculate gross profit and gross margin for both years with the formulas above. Check: Year A = $8,000 and 40%; Year B = $7,700 and 35%. Compare your answers with AI, then correct any mismatch yourself.

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- In pairs, write a short briefing for a store manager. Explain that revenue increased by $2,000 while gross profit decreased by $300 and margin fell by 5 percentage points. Suggest one additional piece of information to investigate; do not claim to know the cause.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W9_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- Introduction: Explain how sales can rise while gross profit falls in this case.
- Method: Copy the prompt you used to compare your calculations with AI and the prompt you used to draft the brief.
- Results: Include the two-row calculation table (gross profit and gross margin for Year A and Year B) and any AI mismatch you corrected.
- Conclusion: Include the 100-word manager brief with facts, an explanation clearly labeled as unverified, and one follow-up question.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- Both calculations are correct, gross margin is not mislabeled as net profit, and the brief does not present a guess as a fact.